# Stat 110 Strategic Practice 8 & Homework 8: Solutions

Prof. Joe Blitzstein (Department of Statistics, Harvard University)

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## Strategic Practice 8 Solutions

### 1. Covariance and Correlation

**Problem 1.** $X$ and $Y$ are outcomes of a green and orange fair die, independent.

**(a)** Using bilinearity of covariance,

$$\text{Cov}(X+Y,\, X-Y) = \text{Cov}(X,X) - \text{Cov}(X,Y) + \text{Cov}(Y,X) - \text{Cov}(Y,Y) = \text{Var}(X) - \text{Var}(Y) = 0,$$

since $X$ and $Y$ are i.i.d. and $\text{Cov}(X,Y) = 0$ by independence.

**(b)** $X+Y$ and $X-Y$ are not independent, even though they are uncorrelated. If $X+Y = 12$, then the only possibility is $X = Y = 6$, forcing $X - Y = 0$. So $P(X-Y=0 \mid X+Y=12) = 1$, which is far larger than $P(X-Y=0) = 6/36 = 1/6$. As another angle: $X+Y$ and $X-Y$ must have the same parity (since their difference $2Y$ is always even), so knowing $X+Y$ is odd immediately tells us $X-Y$ is odd too.

**Problem 2.** In the chicken-egg model, writing $Y = N - X$ for the non-hatching eggs, we know from class that $X \sim \text{Pois}(\lambda p)$ and $Y \sim \text{Pois}(\lambda q)$ are independent, with $N = X + Y$. So

$$\text{Cov}(N,X) = \text{Cov}(X+Y,X) = \text{Var}(X) + \text{Cov}(Y,X) = \lambda p + 0 = \lambda p.$$

Since $\text{SD}(N) = \sqrt{\lambda}$ and $\text{SD}(X) = \sqrt{\lambda p}$,

$$\text{Corr}(N,X) = \frac{\lambda p}{\sqrt{\lambda} \cdot \sqrt{\lambda p}} = \sqrt{p}.$$

**Problem 3.** We want $Z = aX + bY$ and $W = cX + dY$ to be uncorrelated and each standardized (mean 0, variance 1). A natural choice is $Z = X$ (so $a = 1$, $b = 0$). Then

$$\text{Cov}(Z,W) = \text{Cov}(X,\, cX + dY) = c\,\text{Var}(X) + d\,\text{Cov}(X,Y) = c + d\rho = 0,$$

so $c = -d\rho$. Requiring $\text{Var}(W) = 1$:

$$\text{Var}(cX+dY) = c^2 + 2cd\rho + d^2 = d^2(\rho^2 - 2\rho^2 + 1) = d^2(1-\rho^2) = 1,$$

giving $d = 1/\sqrt{1-\rho^2}$ and $c = -\rho/\sqrt{1-\rho^2}$. The solution is

$$a = 1, \quad b = 0, \quad c = \frac{-\rho}{\sqrt{1-\rho^2}}, \quad d = \frac{1}{\sqrt{1-\rho^2}}.$$

**Problem 4.** In the Multinomial story, each of the $n$ objects lands in one of $k$ categories. Let $I_i$ indicate that object $i$ lands in category 1 and $J_i$ indicate that object $i$ lands in category 2. Then $X_1 = \sum_{i=1}^n I_i$ and $X_2 = \sum_{j=1}^n J_j$. By bilinearity,

$$\text{Cov}(X_1, X_2) = \sum_{i,j} \text{Cov}(I_i, J_j).$$

When $i \neq j$, objects $i$ and $j$ are categorized independently, so $\text{Cov}(I_i, J_j) = 0$. The diagonal terms ($i = j$) give

$$\text{Cov}(I_i, J_i) = E(I_i J_i) - E(I_i)E(J_i) = 0 - p_1 p_2 = -p_1 p_2,$$

since $I_i J_i = 0$ always (an object cannot be in two categories at once). Therefore $\text{Cov}(X_1, X_2) = n(-p_1 p_2) = -np_1 p_2$. The same argument gives $\text{Cov}(X_i, X_j) = -np_i p_j$ for all $i \neq j$.

**Problem 5.** The identity $\max(X, Y) + \min(X, Y) = X + Y$ is true as a statement about random variables: for every outcome $\omega$, $\max(X(\omega), Y(\omega)) + \min(X(\omega), Y(\omega)) = X(\omega) + Y(\omega)$. So yes, this always holds.

The covariance claim is incorrect. Covariance is a single number summarizing two random variables globally, not something computed outcome-by-outcome. The r.v. $\max(X, Y)$ is not equal to $X$ as a function; it changes which one it is from trial to trial. In fact, when $X$ and $Y$ are i.i.d. (and therefore uncorrelated), $\text{Cov}(\max(X, Y), \min(X, Y)) > 0$: the max and min are positively correlated because a large minimum forces the maximum to also be large. The argument "sometimes the max is $X$ and sometimes $Y$, and covariance is symmetric" conflates the random variable $\max(X, Y)$ with the pair of ordinary numbers $\{X(\omega), Y(\omega)\}$ at a fixed outcome.

**Problem 6.** Let $X = V + W$, $Y = V + Z$ where $V, W, Z \overset{iid}{\sim} \text{Pois}(\lambda)$.

**(a)**

$$\text{Cov}(X, Y) = \text{Cov}(V + W, V + Z) = \text{Var}(V) + \text{Cov}(V, Z) + \text{Cov}(W, V) + \text{Cov}(W, Z) = \lambda,$$

since $V, W, Z$ are independent and all cross-covariances are zero.

**(b)** $X$ and $Y$ are not independent: $\text{Cov}(X, Y) = \lambda > 0$. Alternatively, $E(Y) = 2\lambda$ but $E(Y \mid X = 0) = \lambda$ (if $X = 0$ then $V = W = 0$, leaving only $Z$), so the distributions differ.

Given $V$, $X = V + W$ and $Y = V + Z$ are functions of $W$ and $Z$ respectively (plus the now-fixed constant $V$), so they are conditionally independent given $V$, since $W$ and $Z$ are independent.

**(c)** Condition on $V$:

$$P(X = x, Y = y) = \sum_{v=0}^{\min(x,y)} P(X = x \mid V = v)P(Y = y \mid V = v)P(V = v)$$

$$= \sum_{v=0}^{\min(x,y)} \frac{e^{-\lambda}\lambda^{x-v}}{(x-v)!} \cdot \frac{e^{-\lambda}\lambda^{y-v}}{(y-v)!} \cdot \frac{e^{-\lambda}\lambda^{v}}{v!} = e^{-3\lambda}\lambda^{x+y} \sum_{v=0}^{\min(x,y)} \frac{\lambda^{-v}}{(x-v)!(y-v)!v!}.$$

The upper limit is $\min(x, y)$ since $V \leq X$ and $V \leq Y$ must both hold.

**Problem 7.** $X$ is Hypergeometric with parameters $w$ (white balls), $b$ (black balls), $n$ (draws), and $N = w + b$.

**(a)** $\binom{X}{2}$ counts the number of pairs of draws that both land on white balls. Each of the $\binom{n}{2}$ pairs of draw positions independently has the same probability of both being white: the probability that the first of the pair is white times the conditional probability that the second is white given the first was, which is $\frac{w}{N} \cdot \frac{w-1}{N-1}$. By linearity,

$$E\binom{X}{2} = \binom{n}{2} \cdot \frac{w}{N} \cdot \frac{w-1}{N-1}.$$

**(b)** Since $E\binom{X}{2} = \frac{E(X^2)-E(X)}{2}$, we get $E(X^2) = 2E\binom{X}{2} + E(X)$. With $p = w/N$, $q = 1 - p$, and $E(X) = np$,

$$E(X^2) = n(n-1)p \cdot \frac{w-1}{N-1} + np.$$

Note that $\frac{w-1}{N-1} = \frac{Np-1}{N-1}$. Then

$$\text{Var}(X) = E(X^2) - (np)^2 = n(n-1)p \cdot \frac{Np-1}{N-1} + np - n^2p^2.$$

Combining terms over a common denominator $N(N-1)$ and simplifying (several intermediate cancellations occur) yields

$$\mathrm{Var}(X) = \frac{N-n}{N-1} \cdot npq.$$

## 2. Transformations

**Problem 1.** $X \sim \mathrm{Unif}(0,1)$.

For $Y = X^2$: set $y = x^2$, so $x = \sqrt{y}$ and $|dx/dy| = 1/(2\sqrt{y})$. For $y \in (0,1)$,

$$f_Y(y) = f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}} = \frac{1}{2} y^{-1/2}.$$

This is the $\mathrm{Beta}(1/2, 1)$ distribution.

For $Y = \sqrt{X}$: set $y = x^{1/2}$, so $x = y^2$ and $|dx/dy| = 2y$. For $y \in (0,1)$,

$$f_Y(y) = f_X(y^2) \cdot 2y = 2y.$$

This is the $\mathrm{Beta}(2,1)$ distribution.

**Problem 2.** $U \sim \mathrm{Unif}(0, 2\pi)$, $T \sim \mathrm{Expo}(1)$ independent. $X = \sqrt{2T}\cos U$, $Y = \sqrt{2T}\sin U$.

Note $X^2 + Y^2 = 2T$ and $U = \arctan(Y/X)$. The Jacobian of the inverse map from $(u,t)$ to $(x,y)$ has absolute determinant 1 (computed as the absolute value of $\det[\partial(x,y)/\partial(u,t)]$, which evaluates to $|-\sin^2 u - \cos^2 u| = 1$). So

$$f_{X,Y}(x,y) = f_{U,T}(u,t) \cdot |J|^{-1} = \frac{1}{2\pi} e^{-t} = \frac{1}{2\pi} e^{-(x^2+y^2)/2} = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2}.$$

This factors into a function of $x$ alone times a function of $y$ alone, so $X$ and $Y$ are independent, each $\sim N(0,1)$. This is the Box-Muller transform for generating standard normal random variables.

**Problem 3.** Let $T = X + Y$ and $W = X$. The transformation $(x,y) \mapsto (t,w) = (x+y, x)$ has Jacobian matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ with absolute determinant 1. Since $y = t - w$,

$$f_{T,W}(t,w) = f_{X,Y}(w,\, t-w) = f_X(w) f_Y(t-w).$$

Integrating out $w$ gives the marginal PDF of $T$:

$$f_T(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t-x)\, dx,$$

the convolution formula.

## 3. Existence

**Problem 1.** Generate $m$ binary strings of length $n$ independently by flipping a fair coin for each bit. Let $S$ be this random set. We want to show $P(S \text{ is } k\text{-complete}) > 0$.

There are $N = \binom{n}{k} 2^k$ conditions to satisfy (one for each choice of $k$ positions and each binary string of length $k$). Let $A_j$ be the event that condition $j$ fails (no string in $S$ matches the required pattern at the required positions). For each condition, a single random string misses it with probability $1 - 2^{-k}$, and the $m$ strings are independent, so $P(A_j) = (1-2^{-k})^m$. By the union bound,

$$P\left(\bigcup_{j=1}^{N} A_j\right) \le N(1-2^{-k})^m = \binom{n}{k} 2^k (1-2^{-k})^m < 1$$

by hypothesis. So $P(S \text{ is } k\text{-complete}) \geq 1 - P(\bigcup A_j) > 0$, meaning a $k$-complete set of size at most $m$ must exist.

**Problem 2.** Pick a pair of students uniformly at random. For each of the 8 problems, let $I_j$ be the indicator that at least one of the two students answered problem $j$ correctly. Since at most 35 students got problem $j$ wrong, the probability that a randomly chosen student missed it is at most $35/100$. By independence of the two students,

$$P(I_j = 0) \leq \left(\frac{35}{100}\right)^2 = 0.1225.$$

By linearity, the expected number of problems that at least one of the pair answers correctly is at least $8(1-0.1225) = 7.02$. Since the expected score exceeds 7, some pair of students must score at least $\lceil 7.02 \rceil = 8$, i.e., they collectively answered every problem correctly.

**Problem 3.** Inscribe a random square in the circle by choosing a uniformly random starting point on the circumference; the four corners are then at angles $0, 90, 180, 270$ from that starting point. Let $I_j$ be the indicator that corner $j$ touches red. By the uniformity of the starting point and since $2/3$ of the circumference is red, $E(I_j) = 2/3$ for each $j$. By linearity, the expected number of red corners is $4 \cdot (2/3) = 8/3 > 2$. Since $8/3$ is not an integer and the number of red corners is always an integer, some inscribed square must have at least 3 red corners.

**Problem 4.** Take a honeycomb tiling of the plane where each hexagonal cell has a circle of the same radius as the coins inscribed in it. Shift the entire tiling by a uniformly random horizontal and vertical offset (the periodicity of the tiling bounds how large these shifts need to be). For each of the 10 designated points, let $I_j$ be the indicator that point $j$ falls inside one of the inscribed circles. By the given geometric fact (the area ratio of inscribed circle to hexagon exceeds 0.9), every point in the plane has probability greater than 0.9 of landing inside an inscribed circle under a uniformly random shift. So $E(I_j) > 0.9$ and

$$E(I_1 + \cdots + I_{10}) > 9.$$

Since the expected number of covered points exceeds 9, there must exist a particular shift for which all 10 points are covered. Placing coins over the circles that contain the points covers all 10 points with no stacking needed (each point is in at most one circle per hexagon cell, and the cells tile without overlap).

# Homework 8 Solutions

## 1. Distinct Birthdays

Let $I_j$ indicate that at least one of the 110 people was born on day $j$ of the year ($j = 1, \ldots, 365$). Then $X = \sum_{j=1}^{365} I_j$ and each $I_j \sim \text{Bern}(p)$ where

$$p = 1 - \left(\frac{364}{365}\right)^{110}.$$

By linearity, $E(X) = 365p \approx 95.08$.

For the variance, using $\text{Var}(X) = \sum_j \text{Var}(I_j) + 2\sum_{i<j} \text{Cov}(I_i, I_j)$:

$$\text{Var}(X) = 365\, p(1-p) + 2\binom{365}{2}\text{Cov}(I_1, I_2).$$

We need $E(I_1 I_2) = P(\text{at least one person born on day 1 and at least one on day 2})$. By inclusion-exclusion on the complement,

$$P(I_1 = 0 \text{ or } I_2 = 0) = 2\left(\frac{364}{365}\right)^{110} - \left(\frac{363}{365}\right)^{110},$$

so $E(I_1 I_2) = 1 - 2(364/365)^{110} + (363/365)^{110}$. Then $\text{Cov}(I_1, I_2) = E(I_1 I_2) - p^2$. Plugging in gives $\text{Var}(X) \approx 10.02$.

## 2. Correlation of the Max and Min of Two Standard Normals

Let $X, Y \overset{iid}{\sim} N(0,1)$, $M = \max(X,Y)$, $L = \min(X,Y)$. From $M + L = X + Y$ and $M - L = |X - Y|$, taking expectations:

$$E(M) + E(L) = 0, \qquad E(M) - E(L) = E|X - Y|.$$

Since $X - Y \sim N(0,2)$, we have $E|X-Y| = \sqrt{2} \cdot E|Z|$ for $Z \sim N(0,1)$, and $E|Z| = \sqrt{2/\pi}$, so $E(M) - E(L) = 2/\sqrt{\pi}$. Thus $E(M) = 1/\sqrt{\pi}$ and $E(L) = -1/\sqrt{\pi}$.

For the covariance: $ML = \frac{(M+L)^2 - (M-L)^2}{4} = \frac{(X+Y)^2 - |X-Y|^2}{4} = XY$. So

$$\mathrm{Cov}(M,L) = E(ML) - E(M)E(L) = E(XY) + \frac{1}{\pi} = 0 + \frac{1}{\pi} = \frac{1}{\pi}.$$

For the variance, take $\mathrm{Var}(M+L) = \mathrm{Var}(X+Y) = 2$ and expand: $\mathrm{Var}(M) + \mathrm{Var}(L) + 2\mathrm{Cov}(M,L) = 2$. By symmetry $\mathrm{Var}(M) = \mathrm{Var}(L) = v$, so $2v + 2/\pi = 2$, giving $v = 1 - 1/\pi$. Therefore

$$\mathrm{Corr}(M,L) = \frac{1/\pi}{1 - 1/\pi} = \frac{1}{\pi - 1}.$$

## 3. Records in the High Jump

Let $I_j$ indicate that jumper $j$ sets a record. Recall $P(I_j = 1) = 1/j$ and the $I_j$ are mutually independent.

**(a)** $\mathrm{Var}(I_j) = \frac{1}{j} - \frac{1}{j^2}$. Since the $I_j$ are independent (hence uncorrelated), the variance of the total number of records among the first $n$ jumpers is

$$\sum_{j=1}^{n} \left( \frac{1}{j} - \frac{1}{j^2} \right).$$

As $n \to \infty$, $\sum 1/j$ diverges while $\sum 1/j^2$ converges (to $\pi^2/6$), so the variance grows without bound.

**(b)** Let $J_j$ indicate a double record at time $j$ (both $j$ and $j-1$ set records), for $2 \le j \le n$. By independence, $P(J_j = 1) = P(I_{j-1} = 1)P(I_j = 1) = \frac{1}{j(j-1)}$. Using the telescoping identity $\frac{1}{j(j-1)} = \frac{1}{j-1} - \frac{1}{j}$,

$$E\left( \sum_{j=2}^{n} J_j \right) = \sum_{j=2}^{n} \frac{1}{j(j-1)} = \sum_{j=2}^{n} \left( \frac{1}{j-1} - \frac{1}{j} \right) = 1 - \frac{1}{n}.$$

As $n \to \infty$ the expected number of double records approaches 1, even though the expected number of single records grows to infinity.

## 4. PDF of $Z^4$ for $Z \sim N(0,1)$

Let $Y = Z^4$. For $y > 0$, $P(Y \le y) = P(Z^4 \le y) = P(|Z| \le y^{1/4}) = 2\Phi(y^{1/4}) - 1$. Differentiating using the chain rule, with $\varphi$ the standard normal PDF,

$$f_Y(y) = 2\varphi(y^{1/4}) \cdot \frac{1}{4} y^{-3/4} = \frac{1}{2\sqrt{2\pi}} y^{-3/4} e^{-y^{1/2}/2}, \quad y > 0.$$

## 5. PDF of a Product

**(a)** Conditioning on $X = x$ (with $x > 0$),

$$P(T \le t) = \int_0^\infty P(Y \le t/x) f_X(x)\, dx = \int_0^\infty F_Y(t/x) f_X(x)\, dx.$$

Differentiating under the integral,

$$f_T(t) = \int_0^\infty \frac{1}{x} f_Y(t/x) f_X(x)\, dx.$$

Jacobno's formula omits the $1/x$ factor. The error arises because replacing the sum of products by the product introduces a nonlinear change of variables that contributes a Jacobian; Jacobno treats probability densities as if they were probabilities, missing this correction.

**(b)** Let $Z = \log T$, $W = \log X$, $V = \log Y$ (all finite since $X, Y > 0$), so $Z = W + V$. The PDFs of $W$ and $V$ are $f_W(w) = f_X(e^w)e^w$ and $f_V(v) = f_Y(e^v)e^v$ (standard change of variables). By the convolution formula for sums,

$$f_Z(z) = \int_{-\infty}^{\infty} f_W(w) f_V(z-w)\, dw = \int_{-\infty}^{\infty} f_X(e^w)e^w f_Y(e^{z-w})e^{z-w}\, dw.$$

Converting back to $T$ via $f_T(t) = f_Z(\log t)/t$ and substituting $x = e^w$,

$$f_T(t) = \int_0^{\infty} f_X(x) f_Y(t/x) \frac{1}{x}\, dx,$$

confirming the result from (a) and showing Jacobno's missing factor of $1/x$.

## 6. Polar Coordinates and Spherical Symmetry

**(a)** Intuitively: since $f(x, y) = g(x^2 + y^2)$ depends only on the distance from the origin, all points at the same distance are equally likely. Knowing the radius $R$ gives no preference to any particular angle, and vice versa, so $R$ and $\theta$ carry independent information.

Formally, the Jacobian of the map $(r, \theta) \mapsto (x, y) = (r\cos\theta, r\sin\theta)$ has absolute determinant $r$. So

$$f_{R,\theta}(r, \theta) = f_{X,Y}(r\cos\theta, r\sin\theta) \cdot r = r \cdot g(r^2), \quad r \geq 0,\ \theta \in [0, 2\pi).$$

This factors as $h(r) \cdot (1/(2\pi))$ (up to normalization), confirming independence with $\theta \sim \text{Unif}(0, 2\pi)$.

**(b)** For $(X, Y)$ uniform on the unit disk, $f_{X,Y} = 1/\pi$ for $x^2 + y^2 \leq 1$. So $g(r^2) = 1/\pi$ and

$$f_{R,\theta}(r, \theta) = \frac{r}{\pi}, \quad 0 \leq r \leq 1,\ \theta \in [0, 2\pi).$$

Marginalizing: $f_R(r) = 2r$ for $r \in [0, 1]$ and $f_\theta(\theta) = 1/(2\pi)$.

**(c)** For $X, Y \overset{iid}{\sim} N(0, 1)$, $f_{X,Y}(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$, so $g(r^2) = \frac{1}{2\pi} e^{-r^2/2}$ and

$$f_{R,\theta}(r, \theta) = \frac{r}{2\pi} e^{-r^2/2}, \quad r \geq 0,\ \theta \in [0, 2\pi).$$

Marginalizing: $f_R(r) = re^{-r^2/2}$ for $r \geq 0$ (the Rayleigh distribution, or equivalently $R^2 \sim \text{Expo}(1/2)$) and $f_\theta(\theta) = 1/(2\pi)$.

## 7. Cliques and Anticliques in a Random Network

Edges between each pair of nodes are independently present with probability 1/2.

**(a)** For each of the $\binom{n}{k}$ subsets of $k$ nodes, let $X_i$ be the indicator that subset $i$ forms a clique. A clique requires all $\binom{k}{2}$ edges to be present, each with probability 1/2, so $P(X_i = 1) = 2^{-\binom{k}{2}}$. By linearity,

$$E(\text{number of } k\text{-cliques}) = \binom{n}{k} 2^{-\binom{k}{2}}.$$

**(b)** Let $T = \sum_{i=1}^{\binom{n}{3}} X_i$ count triangles ($k = 3$). Since $\text{Var}(X_i) = 2^{-3}(1 - 2^{-3}) = 7/64$,

$$\text{Var}(T) = \binom{n}{3} \cdot \frac{7}{64} + 2 \sum_{i<j} \text{Cov}(X_i, X_j).$$

Two triangle-indicator pairs $(X_i, X_j)$ are independent if the two triangles share 0 or 1 nodes (no common edge). If they share exactly 2 nodes (one common edge), then $E(X_i X_j) = 2^{-5}$ (all 5 distinct pairs must

be present) and $E(X_i)E(X_j) = 2^{-6}$, giving $\mathrm{Cov}(X_i, X_j) = 1/64$. The number of pairs of triangles sharing exactly one edge is $\binom{n}{4}\binom{4}{2} = 6\binom{n}{4}$ (choose 4 nodes, then choose which 2 form the shared edge). So

$$\mathrm{Var}(T) = \frac{7}{64}\binom{n}{3} + 2 \cdot 6\binom{n}{4} \cdot \frac{1}{64} = \frac{7}{64}\binom{n}{3} + \frac{3}{16}\binom{n}{4}.$$

**(c)** Let $C$ = number of $k$-cliques and $A$ = number of $k$-anticliques in the random network. By symmetry (anticliques correspond to cliques in the complementary graph, which also has independent fair-coin edges), $E(A) = \binom{n}{k}2^{-\binom{k}{2}} = E(C)$. So

$$E(C + A) = 2\binom{n}{k}2^{-\binom{k}{2}} = \binom{n}{k}2^{1-\binom{k}{2}} < 1$$

by the hypothesis $\binom{n}{k} < 2^{\binom{k}{2}-1}$. Since $E(C + A) < 1$, we have $P(C + A \geq 1) < 1$, which means $P(C + A = 0) > 0$. Therefore there exists a network with no $k$-cliques and no $k$-anticliques.